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How I Can Steal Your Data with Azure Private Endpoints

Otto Gudszent

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Agenda

- Quick overview about Azure networks
- Private Endpoints
- Azure Private Link Service
- Service Endpoints

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Azure global network



Datacenter Azure Meetup TORINO Network

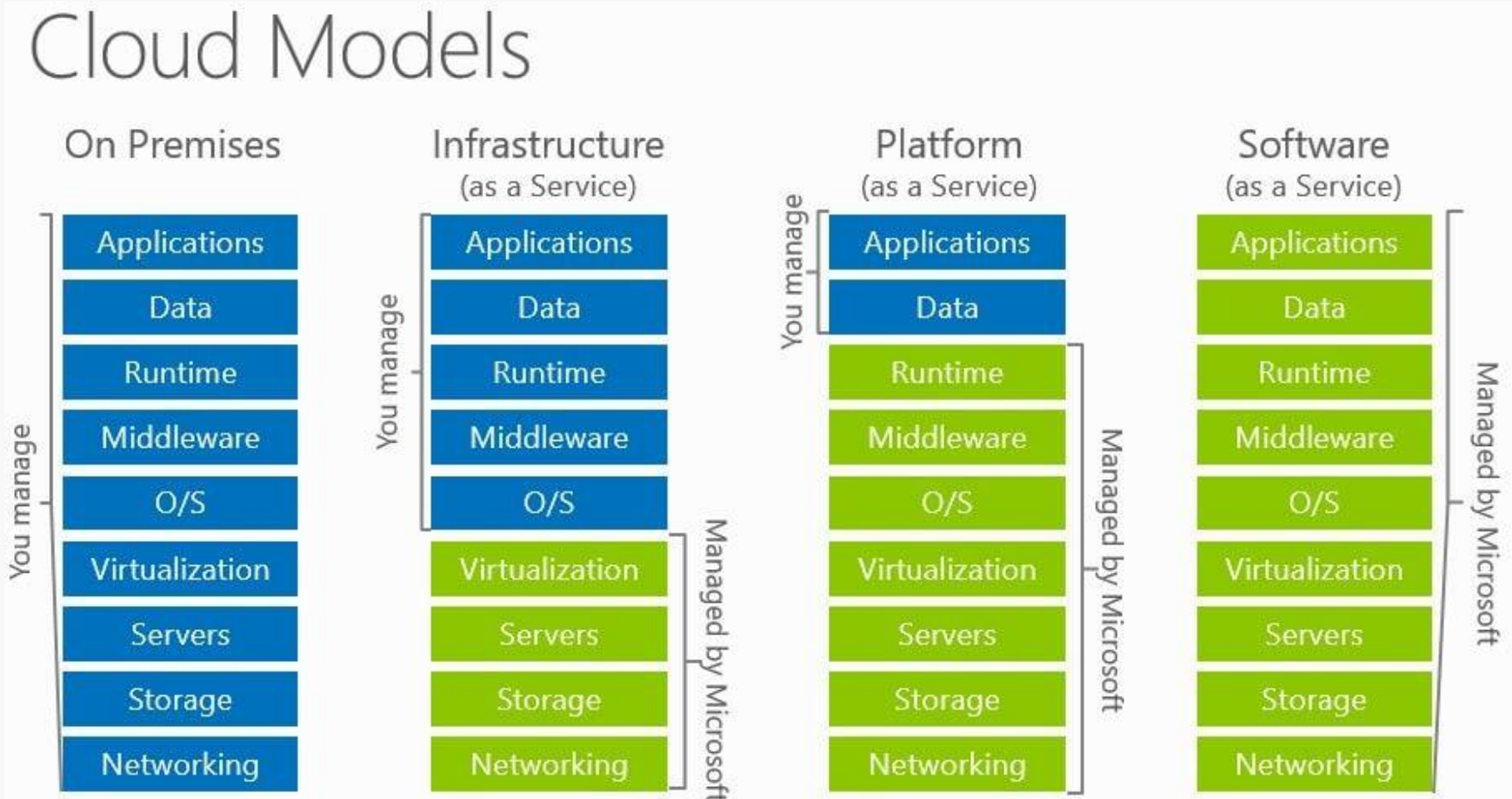
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What Is PaaS in Azure?

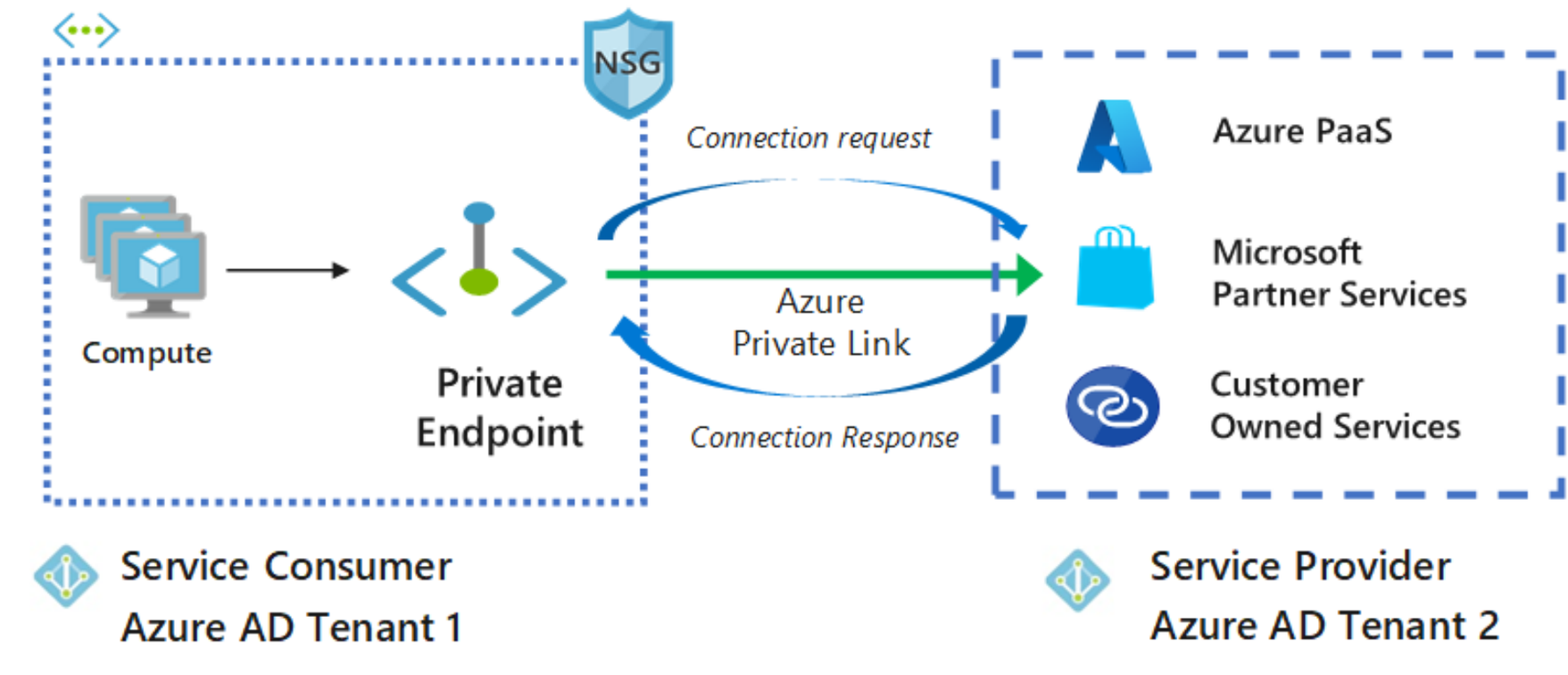
- Platform as a Service: managed services like Storage, SQL, Web Apps
- You focus on your app, Microsoft handles the rest
- Often exposed publicly by default unless locked down



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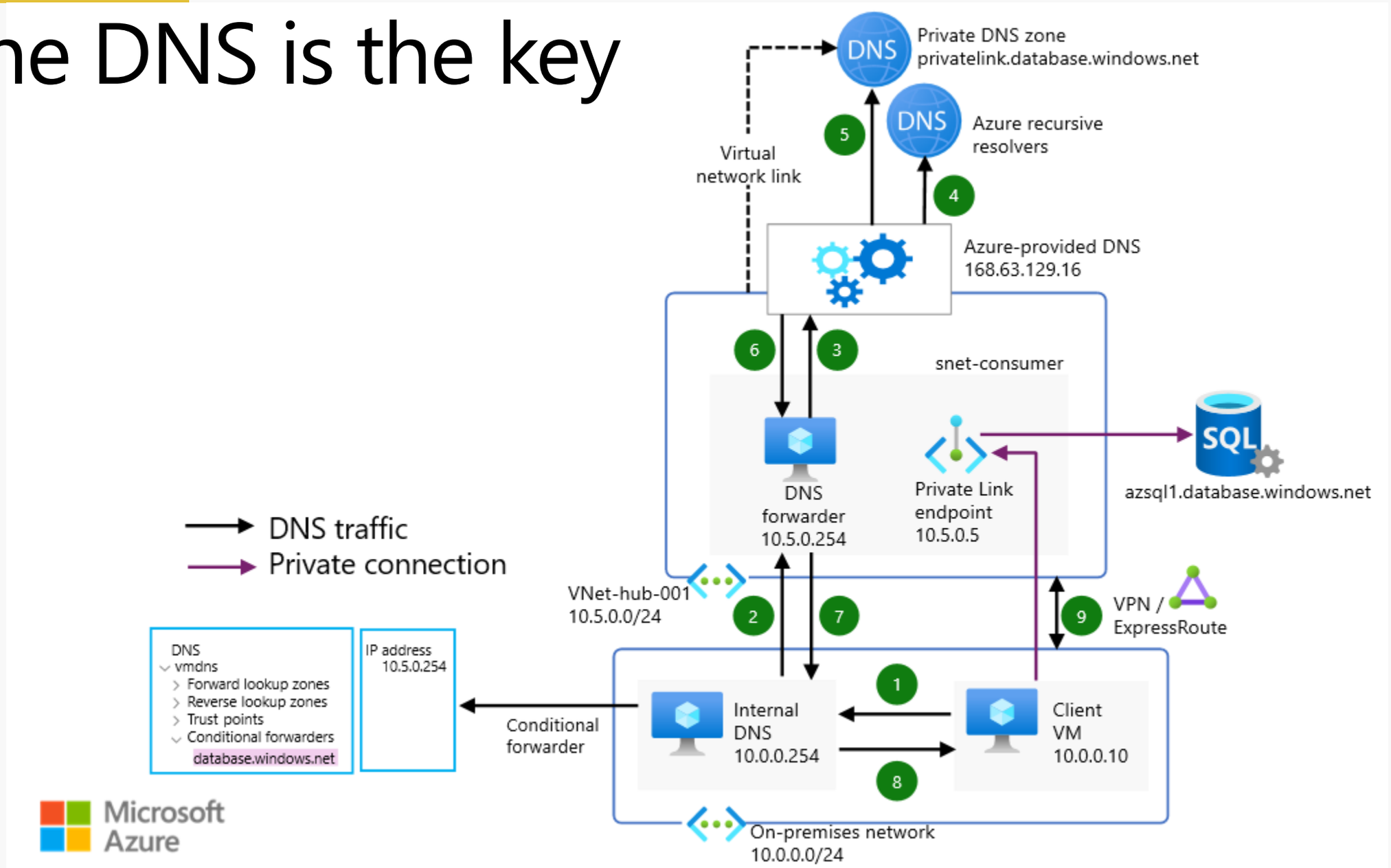
What Are Azure Private Endpoints

- Assign a private IP to a PaaS resource
- Accessible via VNet only, no public internet
- Uses Private Link platform for secure connection

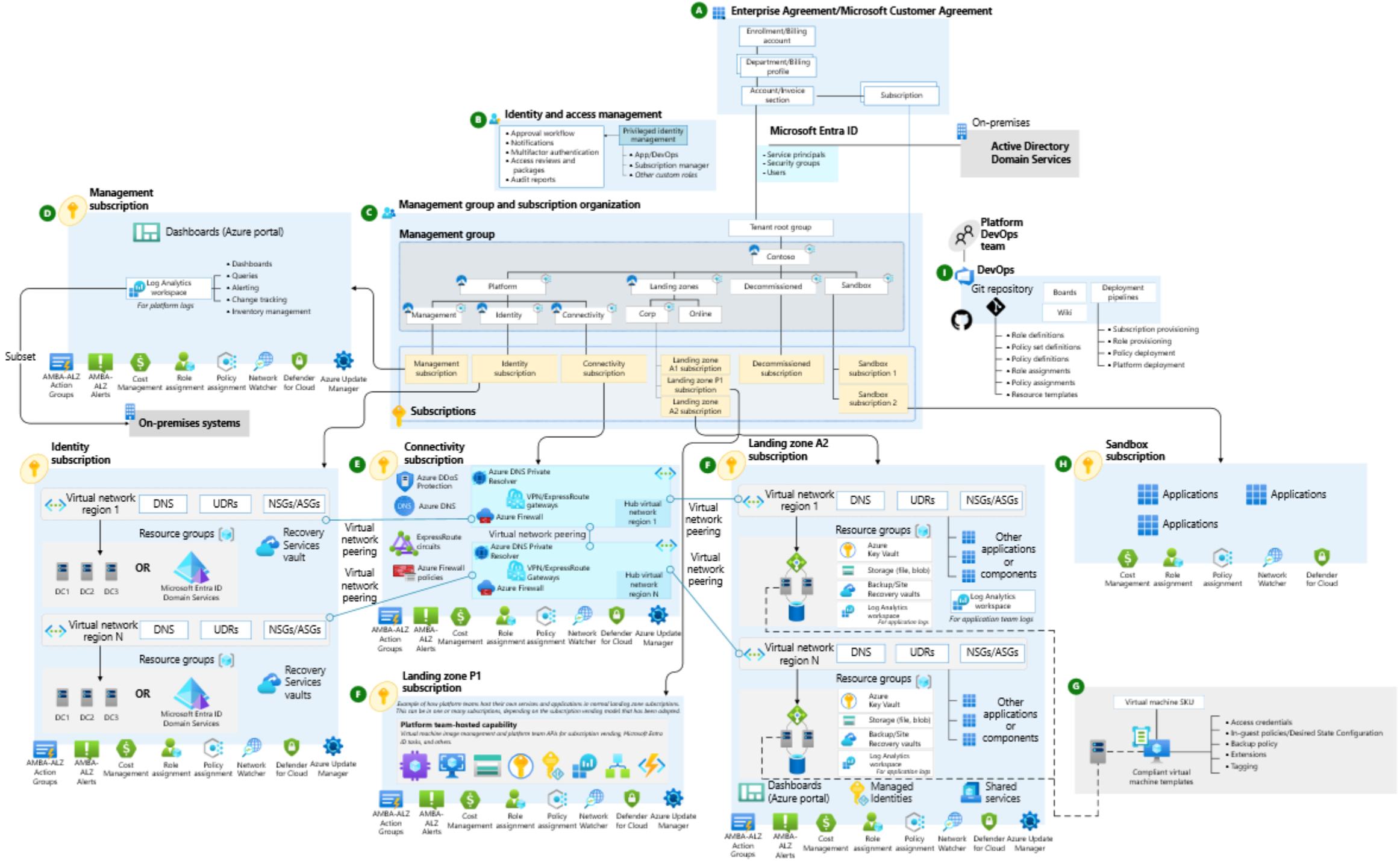


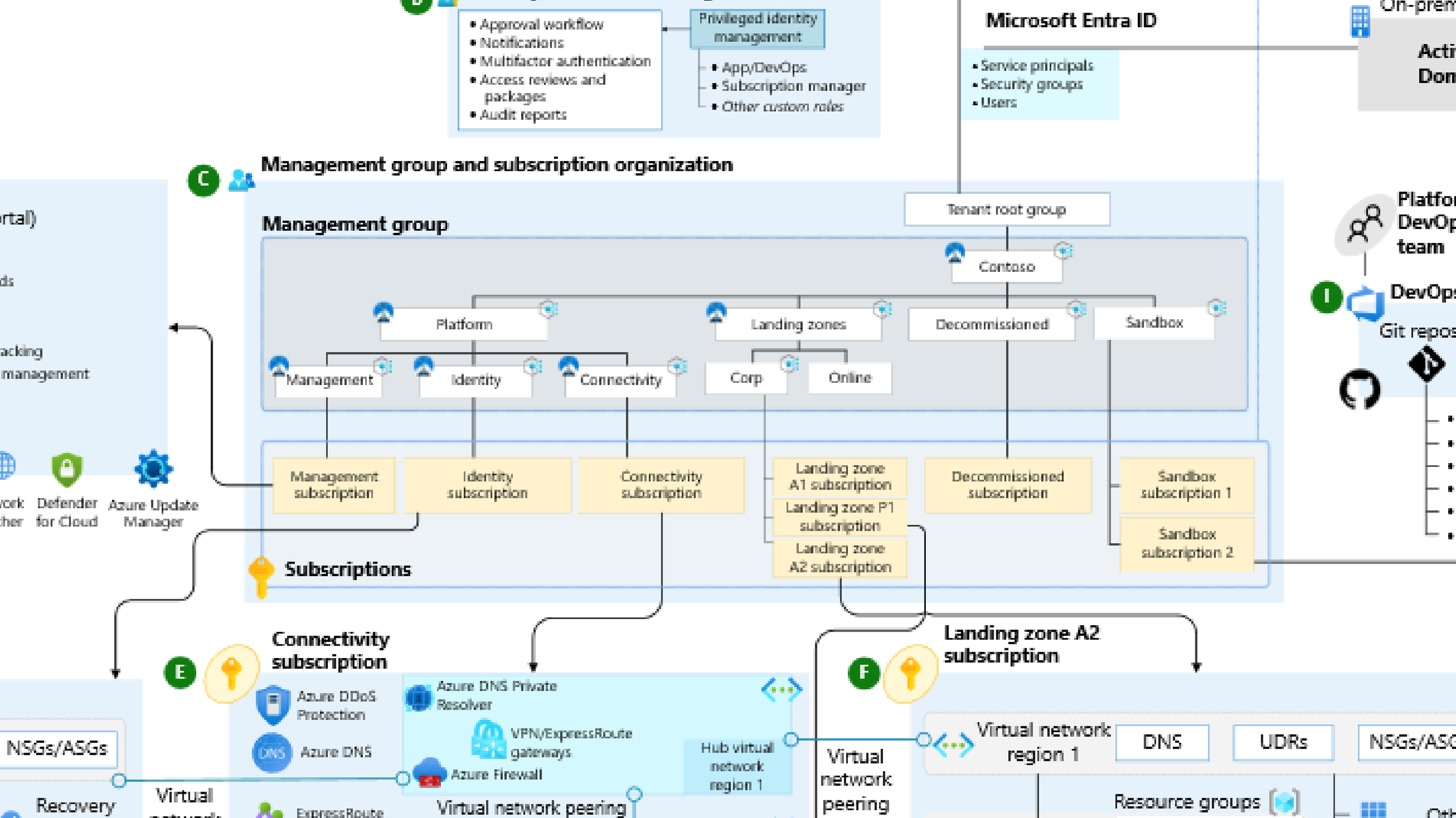
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The DNS is the key



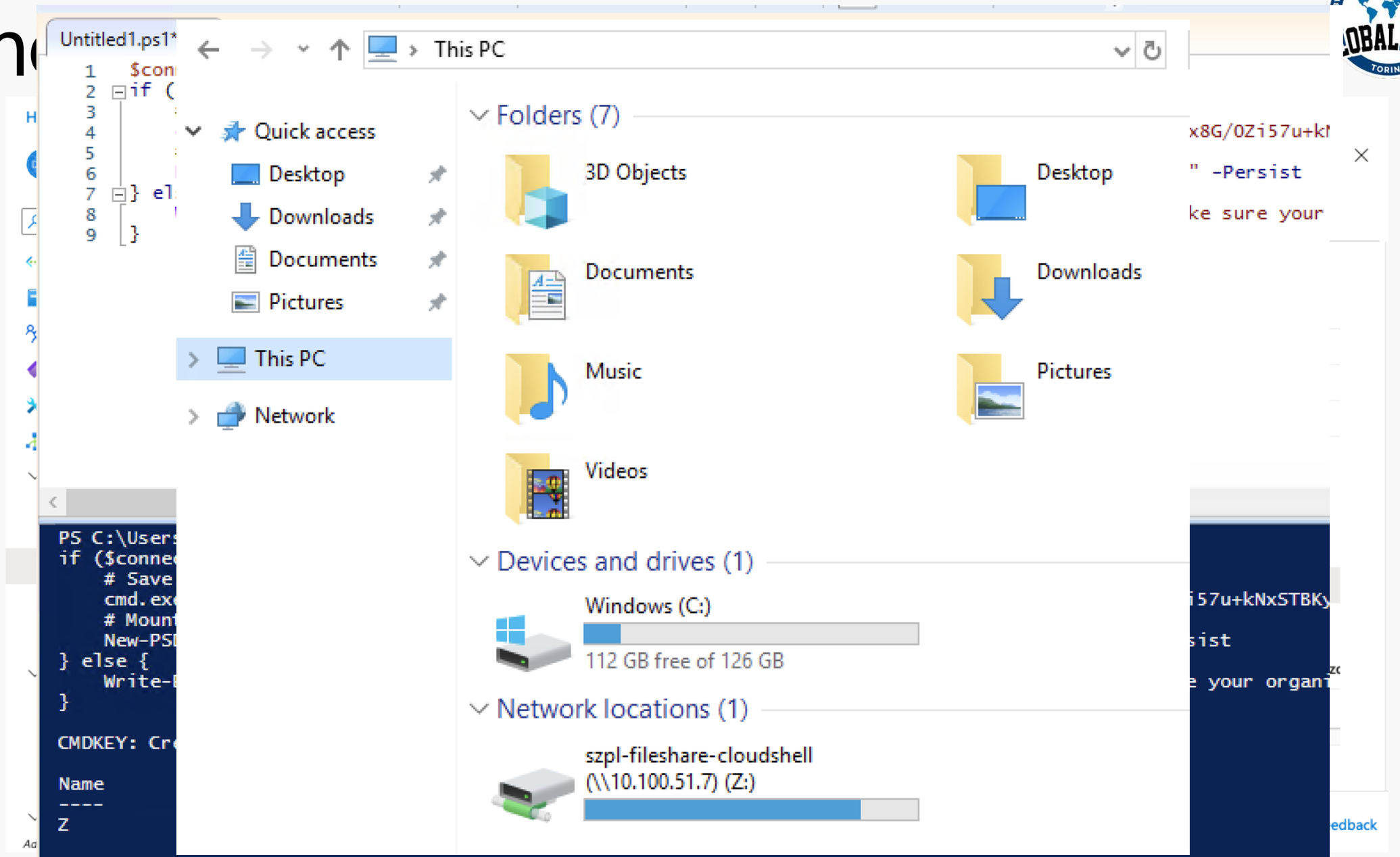
Microsoft Azure


A Enterprise Agreement/Microsoft Customer Agreement
Enrollment/Billing account
Department/Billing profile
Account/Invoice section
Subscription
B Identity and access management
Approval workflow
Notifications
Multifactor authentication
Access reviews and packages
Audit reports
Privileged identity management
App/DevOps
Subscription manager
Other custom roles
Microsoft Entra ID
Service principals
Security groups
Users
On-premises
Active Directory Domain Services
D Management subscription
Dashboards (Azure portal)
Log Analytics workspace
For platform logs
Dashboards
Queries
Alerting
Change tracking
Inventory management
Subset
AMBA-ALZ Action Groups
AMBA-ALZ Alerts
Cost Management
Role assignment
Policy assignment
Network Watcher
Defender for Cloud
Azure Update Manager
On-premises systems
C Management group and subscription organization
Management group
Tenant root group
Contoso
Platform
Landing zones
Decommissioned
Sandbox
Management
Identity
Connectivity
Corp
Online
Management subscription
Identity subscription
Connectivity subscription
Landing zone A1 subscription
Landing zone P1 subscription
Landing zone A2 subscription
Decommissioned subscription
Sandbox subscription 1
Sandbox subscription 2
Subscriptions
Platform DevOps team
I DevOps
Git repository
Boards
Deployment pipelines
Wiki
Role definitions
Policy set definitions
Policy definitions
Role assignments
Policy assignments
Resource templates
Subscription provisioning
Role provisioning
Policy deployment
Platform deployment
Identity subscription
Virtual network region 1
DNS
UDRs
NSGs/ASGs
Resource groups
Recovery Services vault
DC1
DC2
DC3
OR
Microsoft Entra ID Domain Services
Virtual network region N
Recovery Services vaults
E Connectivity subscription
Azure DDoS Protection
Azure DNS
ExpressRoute circuits
Azure Firewall policies
Azure DNS Private Resolver
VPN/ExpressRoute gateways
Azure Firewall
Hub virtual network region 1
Virtual network peering
Hub virtual network region N
F Landing zone A2 subscription
Other applications or components
Azure Key Vault
Storage (file, blob)
Backup/Site Recovery vaults
Log Analytics workspace
For application logs
For application team logs
Dashboards (Azure portal)
Managed Identities
Shared services
F Landing zone P1 subscription
Example of how platform teams host their own services and applications in normal landing zone subscriptions. This can be in one or many subscriptions, depending on the subscription vending model that has been adopted.
Platform team-hosted capability
Virtual machine image management and platform team APIs for subscription vending, Microsoft Entra ID tasks, and others.
H Sandbox subscription
Applications
G
Virtual machine SKU
Compliant virtual machine templates
Access credentials
In-guest policies/Desired State Configuration
Backup policy
Extensions
Tagging


Approval workflow
Notifications
Multifactor authentication
Access reviews and packages
Audit reports
Privileged identity management
App/DevOps
Subscription manager
Other custom roles
Microsoft Entra ID
Service principals
Security groups
Users
On-prem
Acti
Dom
C
Management group and subscription organization
Management group
Tenant root group
Contoso
Platform
Landing zones
Decommissioned
Sandbox
Management
Identity
Connectivity
Corp
Online
Platfo
DevOp
team
I
DevOps
Git repos
Management subscription
Identity subscription
Connectivity subscription
Landing zone A1 subscription
Landing zone P1 subscription
Landing zone A2 subscription
Decommissioned subscription
Sandbox subscription 1
Sandbox subscription 2
Subscriptions
Defender for Cloud
Azure Update Manager
E
Connectivity subscription
Azure DDoS Protection
Azure DNS
Azure DNS Private Resolver
VPN/ExpressRoute gateways
Azure Firewall
Hub virtual network region 1
NSGs/ASGs
Recovery
Virtual
ExpressRoute
Virtual network peering
F
Landing zone A2 subscription
Virtual network region 1
DNS
UDRs
NSGs/ASG
Resource groups
Virtual network peering

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Demo



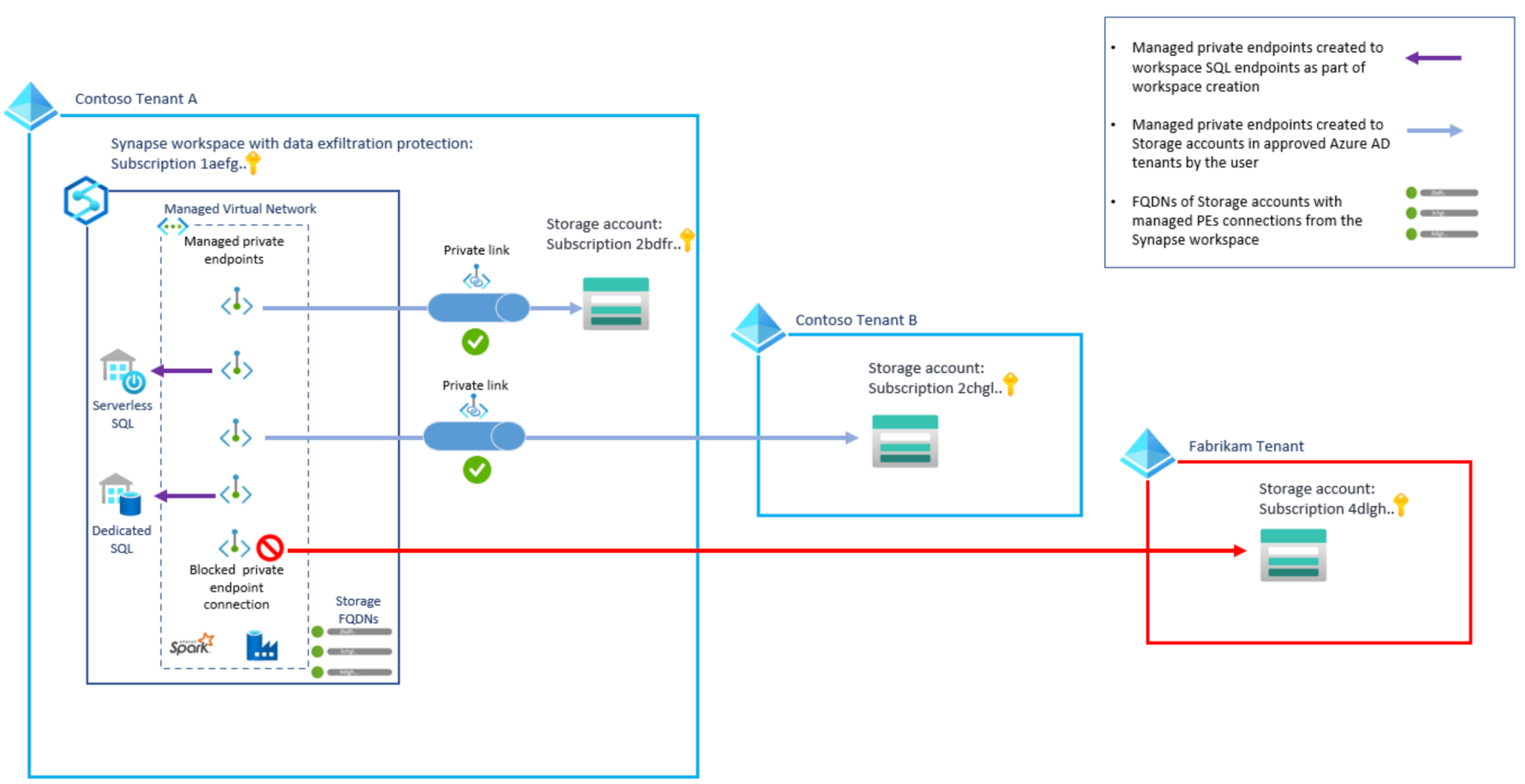
```
"anyOf": [
 {
  "count": {
   "field": "Microsoft.Network/privateEndpoints/manualprivateLinkServiceConnections[*]",
   "where": {
    "allOf": [
     {
      "field": "Microsoft.Network/privateEndpoints/manualprivateLinkServiceConnections[*].privateLinkServiceId",
      "notEquals": ""
     },
     {
      "value": "[split(concat(first(field('Microsoft.Network/privateEndpoints/manualprivateLinkServiceConnections[*].privateLinkServiceId')), '//'), '/')[2]]",
      "notEquals": "[subscription().subscriptionId]"
     }
    ]
   }
  },
  "greaterOrEquals": 1
 },
 {
  "count": {
   "field": "Microsoft.Network/privateEndpoints/privateLinkServiceConnections[*]",
   "where": {
    "allOf": [
     {
      "field": "Microsoft.Network/privateEndpoints/privateLinkServiceConnections[*].privateLinkServiceId",
      "notEquals": ""
     },
     {
      "value": "[split(concat(first(field('Microsoft.Network/privateEndpoints/privateLinkServiceConnections[*].privateLinkServiceId')), '//'), '/')[2]]",
      "notEquals": "[subscription().subscriptionId]"
     }
    ]
   }
  },
  "greaterOrEquals": 1
```

```
"field": "type",
"equals": "Microsoft.Storage/storageAccounts/privateEndpointConnections"


"field": "Microsoft.Storage/storageAccounts/privateEndpointConnections/privateLinkServiceConnectionState.status",
"equals": "Approved"

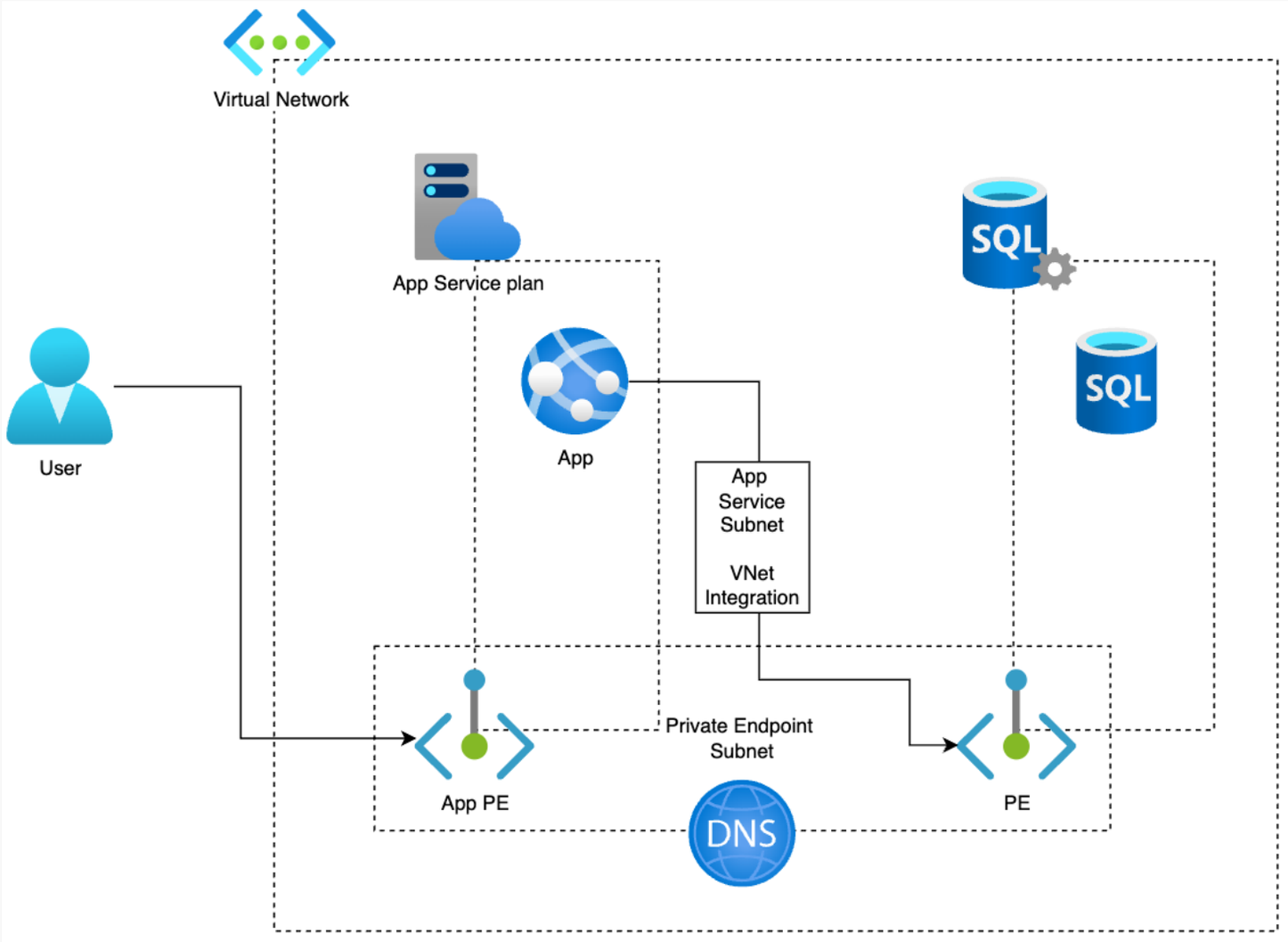

"anyOf": [
    {
        "field": "Microsoft.Storage/storageAccounts/privateEndpointConnections/privateEndpoint.id",
        "exists": false
    },
    {
        "value": "[split(concat(field('Microsoft.Storage/storageAccounts/privateEndpointConnections/privateEndpoint.id'), '//'), '/')[2]]",
        "notEquals": "[subscription().subscriptionId]"
    }
]
```




Contoso Tenant A
Synapse workspace with data exfiltration protection:
Subscription 1aefg..
Managed Virtual Network
Managed private endpoints
Private link
Storage account:
Subscription 2bdfr..
Serverless SQL
Private link
Dedicated SQL
Blocked private endpoint connection
Storage FQDNs
Contoso Tenant B
Storage account:
Subscription 2chgl..
Fabrikam Tenant
Storage account:
Subscription 4dlgh..
Managed private endpoints created to workspace SQL endpoints as part of workspace creation
Managed private endpoints created to Storage accounts in approved Azure AD tenants by the user
FQDNs of Storage accounts with managed PEs connections from the Synapse workspace

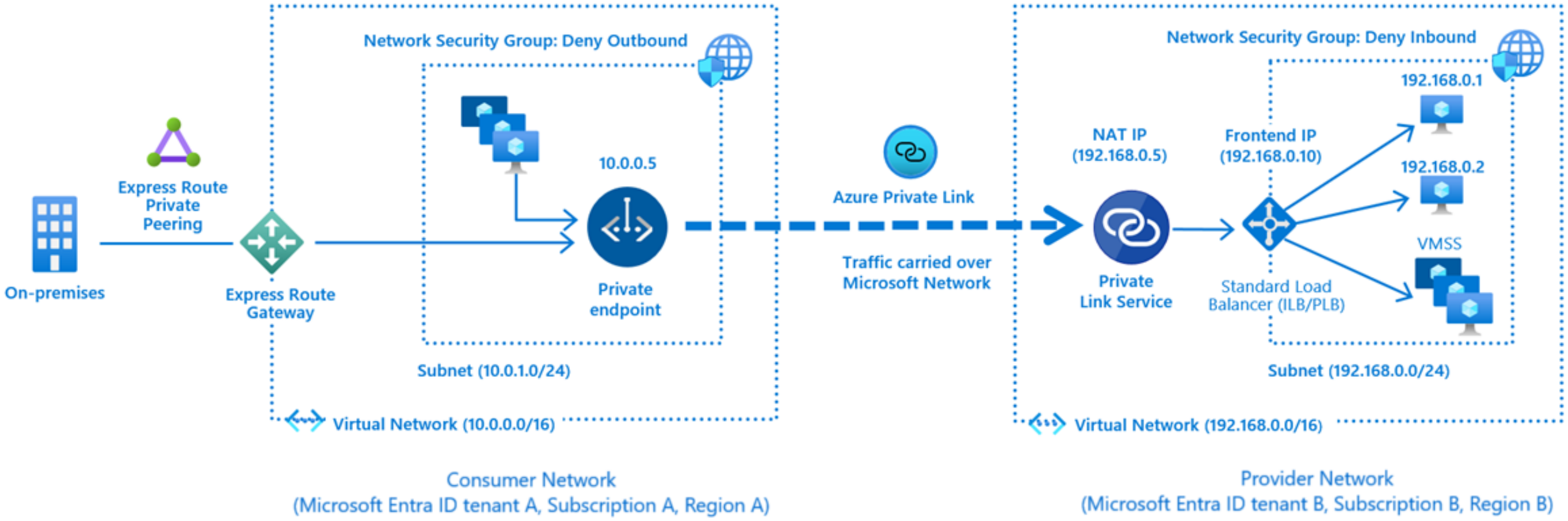
Private Endpoint vs VNET Integration

- Private Endpoint only ingress communication
- VNET Integration can be ingress and egress, depend resources



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Azure Private Link Service



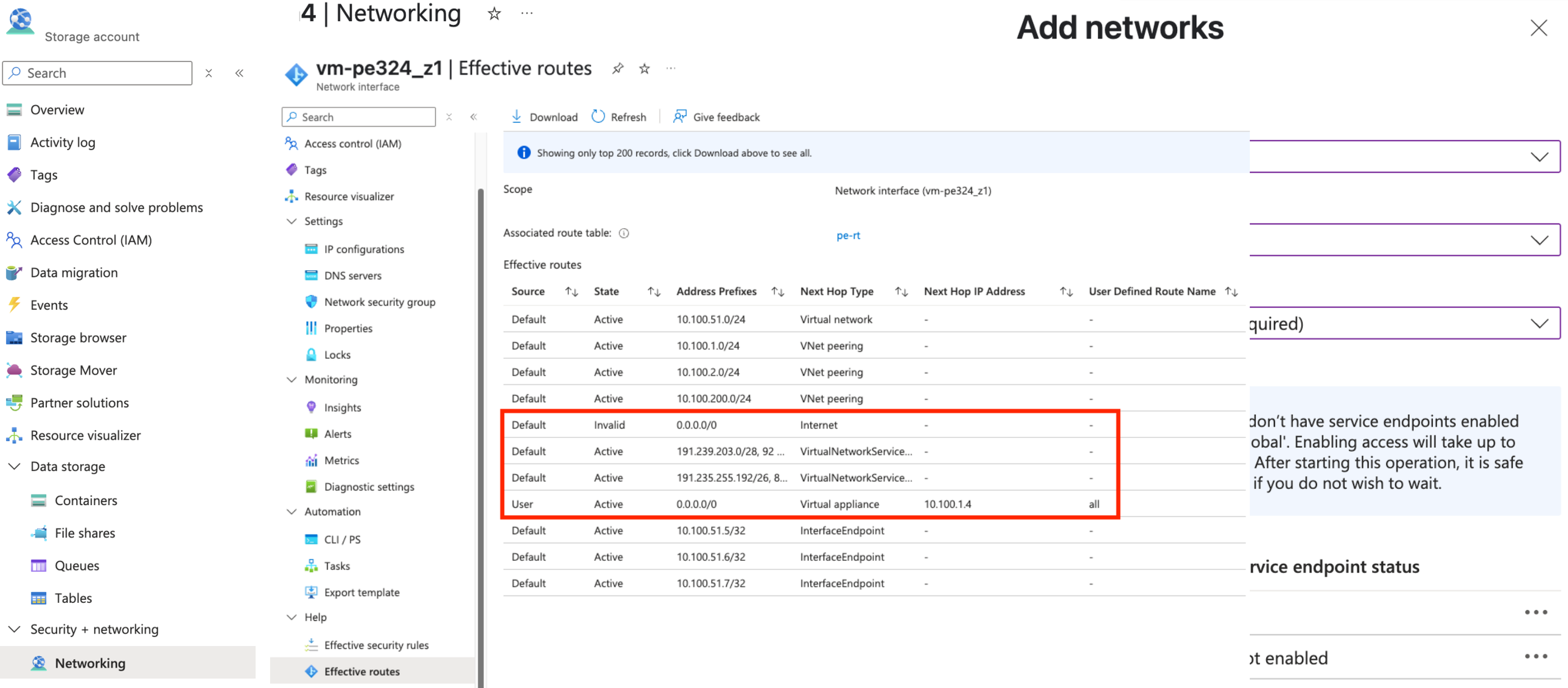
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Service Endpoints

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Virtual network access to another one



Network interface

Search

Access control (IAM)
Tags
Resource visualizer
Settings
IP configurations
DNS servers
Network security group
Properties
Locks
Monitoring
Insights
Alerts
Metrics
Diagnostic settings
Automation
CLI / PS
Tasks
Export template
Help
Effective security rules
Effective routes

Download | Refresh | Give feedback

Showing only top 200 records, click Download above to see all.

Scope: Network interface (vm-pe324_z1)

Associated route table: pe-rt

Effective routes

Source	State	Address Prefixes	Next Hop Type	Next Hop IP Address	User Defined Route Name
Default	Active	10.100.51.0/24	Virtual network	-	-
Default	Active	10.100.1.0/24	VNet peering	-	-
Default	Active	10.100.2.0/24	VNet peering	-	-
Default	Active	10.100.200.0/24	VNet peering	-	-
Default	Invalid	0.0.0.0/0	Internet	-	-
Default	Active	191.239.203.0/28, 92 ...	VirtualNetworkService...	-	-
Default	Active	191.235.255.192/26, 8...	VirtualNetworkService...	-	-
User	Active	0.0.0.0/0	Virtual appliance	10.100.1.4	all
Default	Active	10.100.51.5/32	InterfaceEndpoint	-	-
Default	Active	10.100.51.6/32	InterfaceEndpoint	-	-
Default	Active	10.100.51.7/32	InterfaceEndpoint	-	-

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